

Name:

1. **Production Technologies** Suppose there are two goods produces $(y_1, y_2) \in \mathbb{R}_+^2$ by inputs $(x_1, x_2, x_3) \in \mathbb{R}_+^3$.

(a) Consider the production function $f(x_1, x_2, x_3) = (\overbrace{2x_1}^{y_1} + 4\overbrace{x_2}^{y_2}, \overbrace{x_2}^{y_2} + 2x_3)$. When factor prices are $(w_1, w_2, w_3) = (3, 2, 4)$, graph the set of outputs that costs less than 20. (E.g. $\{(y_1, y_2) \in \mathbb{R}_+^2 \mid w_1 \cdot x \leq 20 \text{ and } y = f(x)\}$) Be sure to label axis, slopes, and important points.

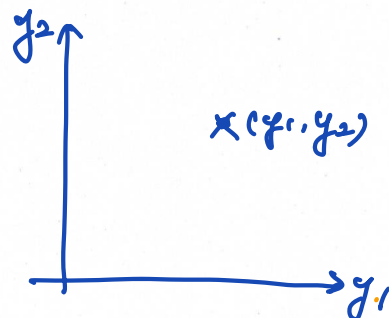
parametric equation - Do NOT TRY THIS
Just out of curiosity!

$$\begin{cases} y_1 = 2x_1 + 4x_2 & \checkmark \leftarrow ① \\ y_2 = x_2 + 2x_3 & \checkmark \leftarrow ② \end{cases}$$

$$3x_1 + 2x_2 + 4x_3 \leq 20$$

$$\begin{cases} 0 \leq x_1 \leq \frac{20}{3} \\ 0 \leq x_2 \leq 10 \\ 0 \leq x_3 \leq 5 \end{cases}$$

$(y_1, y_2) \in \mathbb{R}_+^2$



$$\begin{cases} ①: x_1 = \frac{1}{2}y_1 - 2x_2 \\ ②: x_3 = \frac{1}{2}y_2 - \frac{1}{2}x_2 \end{cases}$$

$$3\left(\frac{1}{2}y_1 - 2x_2\right) + 2x_2 + 4\left(\frac{1}{2}y_2 - \frac{1}{2}x_2\right) \leq 20 \quad \checkmark$$

$$\Leftrightarrow x_2 \geq \frac{1}{4}y_1 + \frac{1}{3}y_2 - \frac{10}{3} \quad \checkmark$$

x_1, x_3

$$x_1 = \frac{1}{2}y_1 - 2x_2 \leq \frac{1}{2}y_1 - 2\left(\frac{1}{4}y_1 + \frac{1}{3}y_2 - \frac{10}{3}\right)$$

$$x_1 = -\frac{2}{3}y_2 + \frac{20}{3} \quad \checkmark = -\frac{2}{3}y_2 + \frac{20}{3} \quad \textcircled{a} \quad 0 \sim \frac{20}{3}$$

$$x_3 = \frac{1}{2}y_2 - \frac{1}{2}x_2 \leq -\frac{1}{8}y_1 + \frac{1}{3}y_2 + \frac{5}{3}$$

⑥

0 ~ 5

$$3x_1 + 2x_2 + 4x_3 = 20$$

$$\textcircled{a} = 0 : y_2 = 10$$

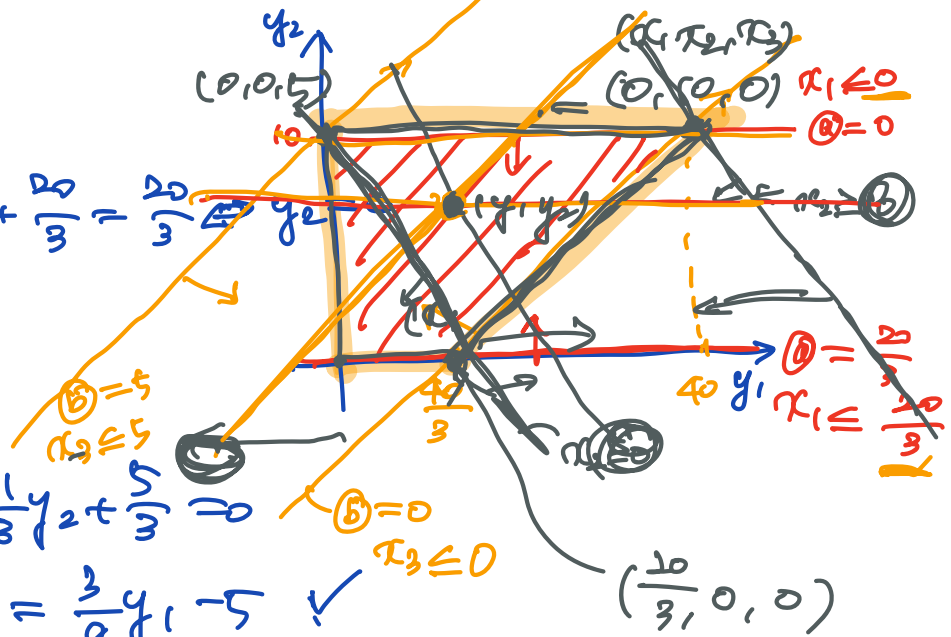
$$\textcircled{a} = \frac{20}{3} : -\frac{2}{3}y_2 + \frac{20}{3} = \frac{20}{3} \Leftrightarrow y_2 = 0$$

$$\textcircled{b} = 0 : -\frac{1}{8}y_1 + \frac{1}{3}y_2 + \frac{5}{3} = 0$$

$$\Leftrightarrow y_2 = \frac{3}{8}y_1 - 5 \quad \checkmark$$

$$\textcircled{b} = 5 : -\frac{1}{8}y_1 + \frac{1}{3}y_2 + \frac{5}{3} = 5$$

$$\Leftrightarrow y_2 = \frac{3}{8}y_1 + 10$$



- ✓
- (b) Consider the production function $f(x_1, x_2, x_3) = (2x_1 + 4x_2, \min\{x_2, 2x_3\})$. When factor prices are $(w_1, w_2, w_3) = (3, 2, 4)$, graph the set of outputs that costs less than 20. (E.g. $\{(y_1, y_2) \in \mathbb{R}_+^2 \mid w_1 \cdot x \leq 20 \text{ and } y = f(x)\}$) Be sure to label axis, slopes, and important points.

$$3x_1 + 2x_2 + 4x_3 \leq 20$$

Extreme points:

$$(0, 10, 0) \Rightarrow (y_1, y_2) = (40, 0)$$

$$(\frac{20}{3}, 0, 0) \Rightarrow (\frac{40}{3}, 0)$$

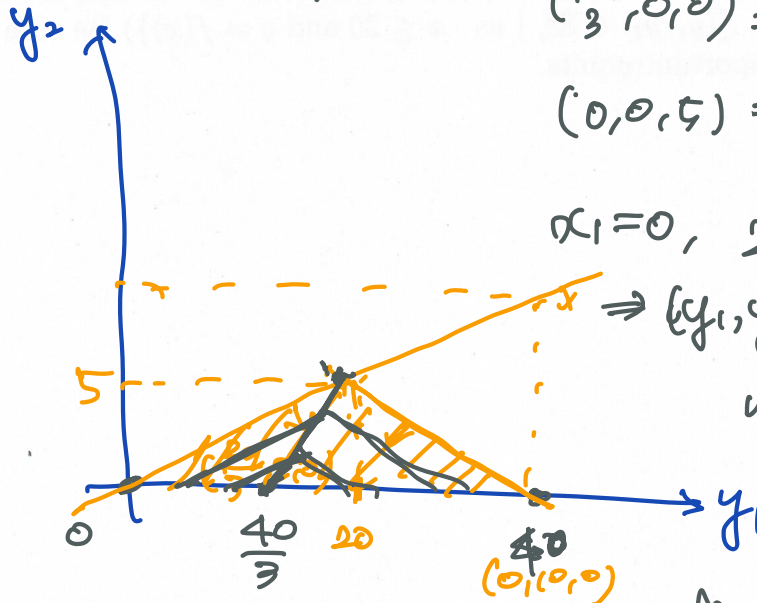
$$(0, 0, 5) \Rightarrow (0, 0)$$

$$\Leftrightarrow x_2 + 2x_3 \leq 10$$

$$x_1 = 0, \quad 2x_2 + 4x_3 \leq 20$$

$$\Rightarrow (y_1, y_2) = (4x_2, x_2)$$

$$\text{where } x_2 \leq 5$$



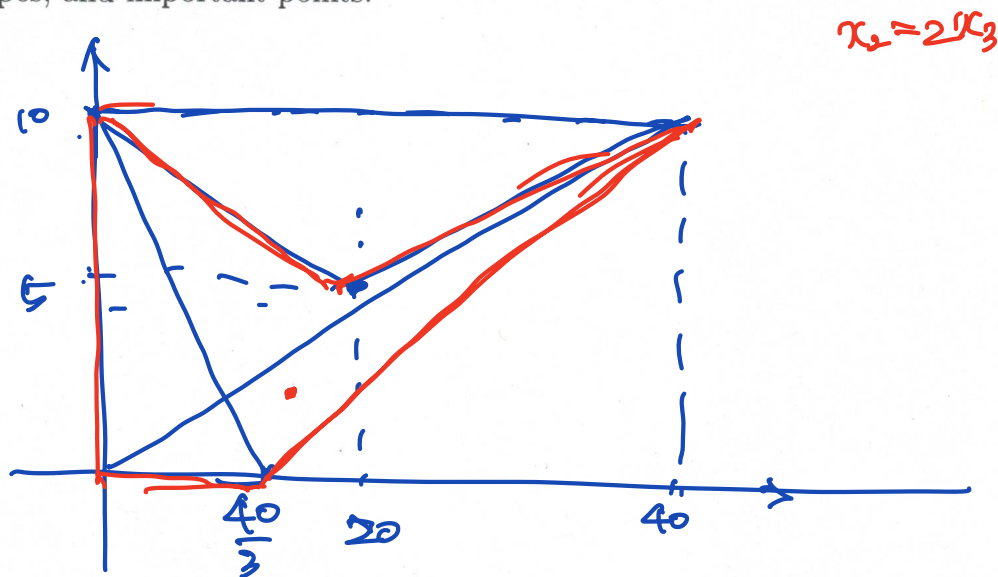
$$\text{If } x_2 > 5, \\ \Leftrightarrow 2x_3 \leq 5$$

$$(y_1, y_2) = (4x_2, 2x_3)$$

$$x_1 \geq 0$$

$$x_1 \uparrow \quad 2x_2 + 4x_3 \leq 20 - x_1$$

- (c) Consider the production function $f(x_1, x_2, x_3) = (2x_1 + 4x_2, \max\{x_2, 2x_3\})$. When factor prices are $(w_1, w_2, w_3) = (3, 2, 4)$, graph the set of outputs that costs less than 20. (E.g. $\{(y_1, y_2) \in \mathbb{R}_+^2 \mid w_1 \cdot x \leq 20 \text{ and } y = f(x)\}$) Be sure to label axis, slopes, and important points.



2. **Cost Minimization** Consider the production function $f(x) = x_1 + x_2^{\alpha_2} x_3^{\alpha_3}$ for $\alpha \in \mathbb{R}_{++}^2$ when factor prices are given by (w_1, w_2, w_3) .

(a) Solve for the short run cost function when $x_1 = \bar{x}_1$.



$\min_{x_2, x_3}$

$$3\bar{x}_1 + 6x_2 + 2x_3$$

$$\text{s.t. } \bar{x}_1 + x_2^{\frac{1}{2}} x_3^{\frac{1}{2}} \leq \bar{y}$$

$\Leftrightarrow \min_{x_2, x_3}$

$$6x_2 + 2x_3$$

$$\text{s.t. } \bar{x}_1 + x_2^{\frac{1}{2}} x_3^{\frac{1}{2}} \leq \bar{y}$$

$$\bar{y} - \bar{x}_1 = \tilde{y}$$

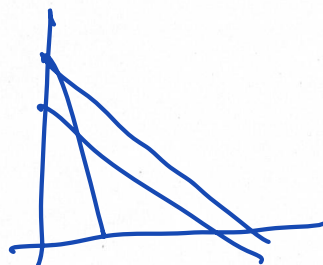
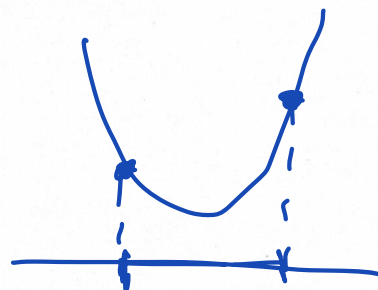
$$L = \underline{6x_2 + 2x_3} - \lambda [\underline{x_2^{\frac{1}{2}} x_3^{\frac{1}{2}} - \bar{y} - \bar{x}_1}] - \tilde{y}$$

- (b) Solve for the short run cost function when $x_2 = \bar{x}_2$. [Hint: the cost function may depend on the parameters.]

cases

- i) ~~all~~ all on x_1
- ii) all on x_3
- iii) mix ←

$$\text{cost} = \min \{ \text{i) ii) iii) } \}$$



(c) Solve for the long run cost function.

$$\hookrightarrow \min \{ (a), (b) \}$$

- (d) When $(w_1, w_2, w_3) = (3, 6, 2)$ and $\alpha = (2, \frac{1}{2})$, graph the short run cost function as a function of $\bar{y}$ when $\bar{x}_1$ is 5, 10, and 20 respectively. Also graph the long run cost curve. How do these two compare?

